

Prompt Confusion Mechanistic Slice — Grouped Update

Update to the earlier brief after rerunning the probe analysis with grouped cross-validation by `matched_pair_id` where available.

DATE	MODEL	CAPTURE RUN	STATUS
April 2026	Qwen3-30B-A3B	09036218e3ae	Grouped rerun complete

BOTTOM LINE

The grouped rerun did *not* knock down the conflict-vs-non-conflict result. That means the earlier concern about pair leakage was not the main issue for that task. The conflict probe still looks almost perfectly separable, which shifts suspicion toward broader prompt-family or task-construction differences. The source probe remains strong, but the grouped rerun did not really test a harder split there because each row was already its own group.

CONFLICT GROUPS	SOURCE GROUPS	BEST CONFLICT LAYER	BEST SOURCE LAYER
972	666	L12	L20
1,944 rows; paired split applied	666 rows; one row per group	Grouped balanced acc 0.996	Grouped balanced acc 0.921

What Changed

Task	Old Best Bal	Grouped Best Bal	Interpretation
Conflict vs non-conflict	0.9979	0.9959	Essentially unchanged; pair leakage was not the main driver.
Strategy vs setting	0.9058	0.9213	Nominally improved, but this grouped split is not materially stronger because groups = rows.

KEY UPDATE

The first rerun answered one question clearly: keeping matched pairs together does *not* explain away the conflict readout. The second rerun answered a more awkward question: using `matched_pair_id` as the group key does almost nothing for the source task, because the strong-conflict-only slice contains one row per pair.

Probe Readout After Grouping

Conflict Vs Non-Conflict

Grouped evaluation still gives near-perfect separation. The best layer stays at **L12**, with balanced accuracy **0.9959**. The weakest layer is still very high at **0.9717**. That is

Strategy Vs Setting

The source probe remains strong, with grouped balanced accuracy peaking at **0.9213** on **L20**. But the grouping key is not doing real work here: the export reports **666**

the real update: the coarse conflict task remains trivially linearly separable even when aligned and strong-conflict rows from the same pair are forced into the same fold.

groups for 666 rows. So this rerun is still informative about signal strength, not yet a tougher generalization test.

Layer	Grouped Bal	Old Bal	Delta
12	0.9959	0.9979	-0.0020
40	0.9943	0.9841	+0.0102
8	0.9933	0.9856	+0.0077
0	0.9717	0.9666	+0.0051

Grouped conflict probe: still too easy. The leakage story has moved up a level, from matched-pair leakage to likely family/template/task leakage.

Layer	Grouped Bal	Old Bal	Delta
20	0.9213	0.8875	+0.0338
4	0.9173	0.9015	+0.0158
12	0.9150	0.8983	+0.0167
0	0.8751	0.8866	-0.0115

Grouped source probe: good signal, but not yet a stricter control because **matched_pair_id** is one-to-one with rows in this slice.

What This Means Now

REVISED TAKE

The pair-grouping fix was necessary, but it did not change the main picture. The conflict task is still suspiciously easy. That means the next control should not be another generic rerun of the same setup; it should be a more meaningful split, such as leaving out a whole family, template, or pressure bucket. For the source task, the current result is encouraging, but it still needs a grouping key that actually bundles related rows together.

Question	Answer After Grouped Rerun
Did matched-pair leakage explain the conflict result?	No. The grouped conflict probe remained near-perfect.
Did the grouped rerun make the source test meaningfully stricter?	Not yet. Each source row was already its own group under matched_pair_id .
What is the next real control?	Leave out broader shared structure: family, template, or pressure bucket; or define a better group key for the source slice.

Recommended Next Step

The next pass should focus on *harder generalization*, not more of the same:

1. For conflict vs non-conflict, try leave-one-family-out or leave-one-template-out probes.
2. For strategy vs setting, define a better grouping key than **matched_pair_id** for the strong-conflict-only slice.
3. Only after one of those survives should we treat the readout as strong enough to motivate finer localization or causal work.

This update supersedes the earlier interpretation of the grouped rerun, but does not replace the earlier brief file.